

- **Problem Statement ID - 26207**

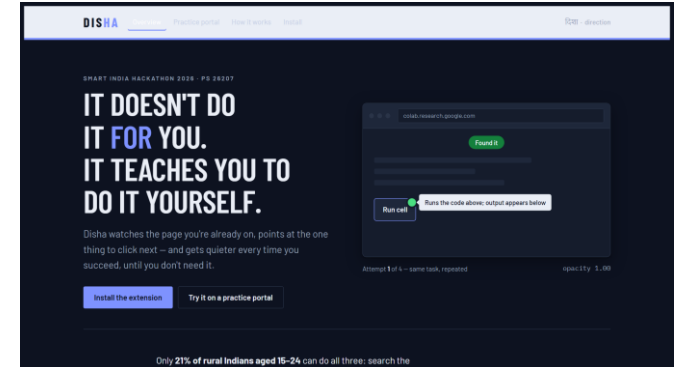
Problem Statement Title - Student Innovation (bring-your-own idea)

Theme - Smart Education

PS Category - Software

Team ID - *issued after the intercollege round*

Team Name - Code Blooded



IDEA / SOLUTION :

Disha reads the page's own controls and their accessible names, then marks the ONE next step with a colour-coded dot.

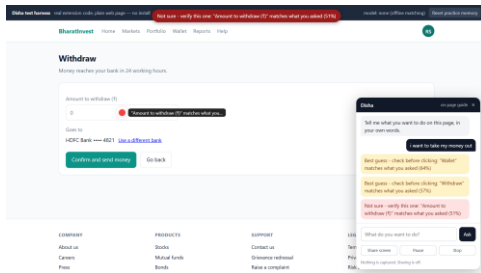
- You click. Disha never can. At an OTP or payment step it pauses and hands control back.
- Every success dims the dot; five clean repetitions remove it entirely.
- Runs on any portal - nothing is scripted in advance.

Problem Resolution :

- A citizen who does not know the word 'withdraw' still finishes the task - and finishes it themselves.
- No paid intermediary, no travel, no 'come back next week'.
- It points instead of clicking, so it is safe where an autopilot agent is not.

Unique Value Propositions (UVP) :

- Confidence you can see - the dot's colour is not the model's self-report; page evidence is scored separately and both must agree.
- It teaches, then leaves - guidance fades on success, returns after a mistake.
- Zero integration cost - no ministry rebuilds anything.



Closest shipped product: Microsoft Copilot Vision "Highlights". We do not claim to have invented on-screen pointing. Ours is the combination - exposed confidence, DOM + ARIA grounding, a fading scaffold, and India-specific task packs.

Technologies used

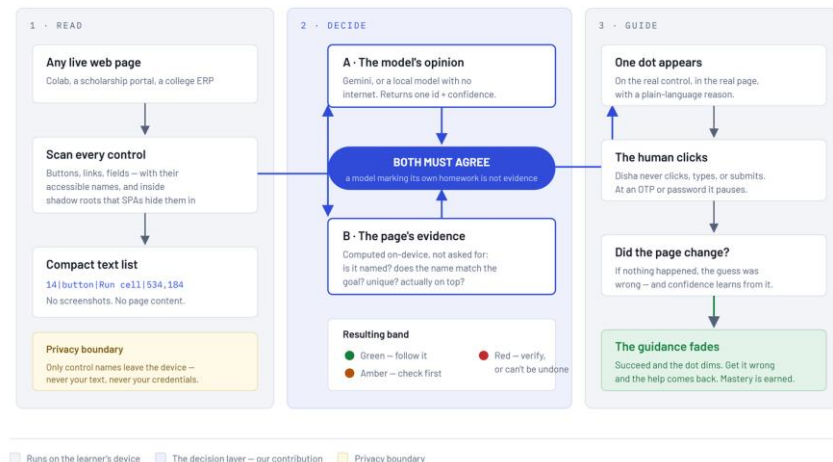
- Manifest V3 Chrome extension, vanilla JavaScript, no build step.
- Model is pluggable - Gemini free tier, Anthropic, or a local Ollama model with no internet.
- DOM + ARIA, not Chrome's accessibility tree - that needs the debugger permission and breaks on old government markup.
- Vision grounding (GUI-Actor, MIT) is the documented fallback.

Methodology and process for implementation

- Green needs 0.80 model AND 0.75 page evidence.
- We display the weaker of the two numbers, never the average - so the number can never contradict the colour.
- **Working prototype, verified end-to-end in a browser.**

HOW DISHA DECIDES WHERE TO POINT

Two independent judgements must agree before the learner is shown a confident marker.



The one thing to take from this diagram: everyone else asks a model where to click and believes the answer. Disha scores the page independently and requires both to agree — which is why its confidence means something, and why it can honestly say "I'm not sure."

page > enumerate interactive elements + ARIA names > compact text list > model returns {id, confidence} > on-device grounding score > colour band > draw dot > human clicks > wait for DOM to settle > next step

Analysis of the feasibility of the idea

Technical, Financial, Market, Operational

- Technical: built and running today; DOM + ARIA works on portals we do not control.
- Financial: a compact text request per step on a free-tier model; local model = zero.
- Operational: installs as an extension. No ministry integration, no portal changes.

Potential challenges and risks

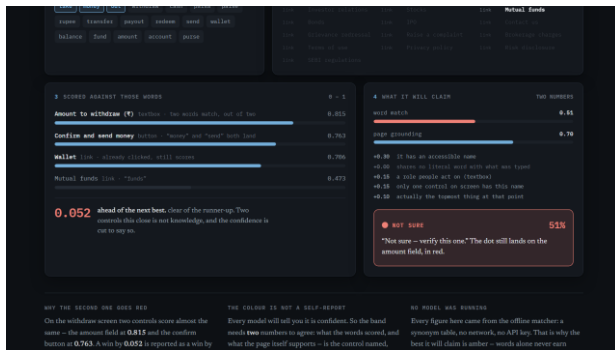
Technical, Financial, Market, Operational

- Unlabelled / icon-only controls - our weakest case.
- Model picks the wrong element.
- Trust: users may over-rely on a confident-looking dot.
- Market: Copilot Vision already points at things.

Strategies for overcoming these challenges

Methods, principle, Strategies, Algorithms

- Unlabelled controls drop the page score, so the dot goes red - never a silent guess; vision grounding is the fallback.
- A wrong pick costs a glance, not a transaction - the human always clicks.
- Over-reliance is answered by the fade itself.



Measured run: "i want to take my money out" produced 64% > 57% > 51% RED > 75% on our practice portal. The amount field scored 0.815 and the confirm button 0.763 - a win by 0.052 is reported as a win by 0.052.

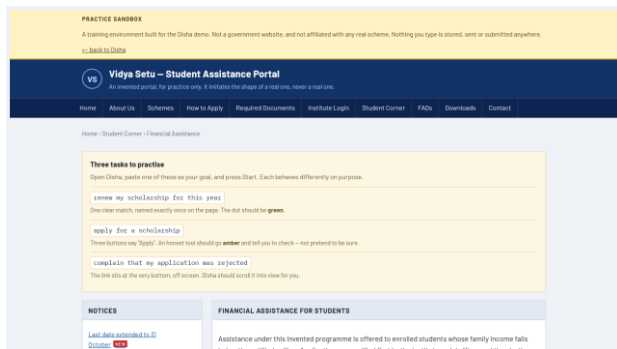
That run used the offline matcher with no model, which is capped below green by design. We have not yet recorded a green dot on stage, and we have not measured accuracy.

Potential impact on the target audience

- Positive - improvement: the citizen completes the task alone instead of paying someone to click.
- New opportunities: a teacher or NGO worker walks a task once and it becomes a pack others reuse.
- Negative - technology adoption: it needs a browser and a first install.

Benefits of the solution (social, economic, environmental, etc.)

- Social - improved access, empowerment: independence on services people are already entitled to.
- Economic - cost: avoids the per-visit intermediary and travel documented in the CSC studies.
- Educational: guidance fades on demonstrated success - active learning +0.47 SD, spaced retrieval $g = 0.74$.



Only 11.4% of connected households use the internet for government services, and one in five households using digital services depends on help from outside the household (NCAER, June 2026). A Maharashtra study recorded Rs 47 in service charges plus travel; a Jharkhand study recorded Rs 101 actually paid against a Rs 30 official fee.

Both learning figures validate the mechanism, not our product. Say that out loud.

Details / Links of the reference and research work

- MoSPI, Comprehensive Annual Modular Survey 2022-23 - the 21% combined ICT-skill figure.
- NSS 78th Round (2020-21) - computer literacy 24.7% national, 18.1% rural.
- NCAER, The Evolving Landscape of Digital Inclusion in India, June 2026.
- PIB / MeitY - CSC network: 48.54 crore transactions, 5,01,731 centres, FY 2025-26.
- Freeman et al., PNAS 2014 - active learning, +0.47 SD across 225 studies.
- Adesope et al. 2017 (retrieval, $g = 0.51$); Latimier et al. 2021 (spacing, $g = 0.74$).
- Renkl et al. - guidance fading triggered by mastery evidence, not elapsed time.
- Microsoft Copilot Vision release notes - the closest shipped prior art.
- IAMAI-Kantar, Internet in India 2024; BHASHINI metrics dashboard.
- microsoft/GUI-Actor (MIT) - the vision-grounding fallback on our roadmap.

Every figure on this deck is traced to an exact source line in deck/numbers.md.